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Time Series Momentum: Can Past Returns Predict the Future?

Replicating Moskowitz, Ooi & Pedersen (2012)

Moskowitz, T.J., Ooi, Y.H. & Pedersen, L.H. (2012). Time Series Momentum. Journal of Financial Economics, 104(2), 228–250.

Topics covered in this lecture:

- Time series momentum (TSMOM): concept, theories, and the paper's core findings
- Building the trend signal: EWMA volatility and the 12-month sign rule
- Statistical validation: pooled predictive regressions across all ETFs and horizons
- Volatility-scaled position sizing and the equal-risk-contribution principle
- Full backtest, comparison to paper, robustness, and honest assessment

Lecture Roadmap

1	**What Is Time Series Momentum?** *The core idea, two types of momentum, and three explanations*	~15 min
2	**Building the Trend Signal** *EWMA volatility estimation and the 12-month sign rule*	~15 min
3	**Statistical Validation** *Pooled predictive regressions for horizons h = 1–24 months*	~10 min
4	**Position Sizing** *Equal risk contribution and the 40% volatility target*	~10 min
5	**Backtesting** *Performance summary, cumulative returns, and drawdowns*	~15 min
6	**Comparing to the Paper** *Alpha grid, TSMOM vs. passive, and the straddle property*	~10 min
7	**Robustness Checks** *Subperiod stability, cost sensitivity, lookback sensitivity*	~10 min
8	**Honest Assessment** *What replicates, what doesn't, and the structural reasons*	~5 min

1

What Is Time Series Momentum?

The core idea, two types of momentum, and three explanations

The Core Idea and Two Types of Momentum

Time Series Momentum — In Plain Terms

- Buy assets that went up over the past 12 months and short assets that went down, sizing each position by volatility. No fundamental analysis — just price direction.

Dimension	Time Series Momentum (this paper)	Cross-Sectional Momentum (Jegadeesh & Titman 1993)
The question	Is THIS asset trending up or down?	Which assets are trending the MOST relative to each other?
Goes long if...	The asset's OWN return was positive	The asset is in the TOP 10% of performers
Goes short if...	The asset's OWN return was negative	The asset is in the BOTTOM 10% of performers
Cares about...	Absolute direction	Relative ranking
Example: Gold +5% vs avg +15%	TS: BUY (own return positive)	XS: SHORT (underperformed peers)

The two strategies can give OPPOSITE signals: Gold up +5% in a market up +15% → TSMOM says BUY; cross-sectional momentum says SHORT.

Why Would Momentum Work? Three Explanations

Theory 1: Underreaction (Behavioral)

- Good news spreads slowly — not all investors react at once. Early buyers push prices up; late buyers continue the trend as they learn. Momentum fades once everyone has reacted.
- Prediction: momentum strongest where information spreads slowly; gradually decays once market fully adjusts.

Theory 2: Overreaction (Behavioral)

- After months of rising prices, investors become overconfident and buy more, pushing prices beyond fundamentals. When reality sets in, prices crash.
- Prediction: continuation (momentum) in short run (1–12 months), then REVERSAL in long run (3–5 years). Paper finds both patterns.

Theory 3: Risk Premium (Risk-Based)

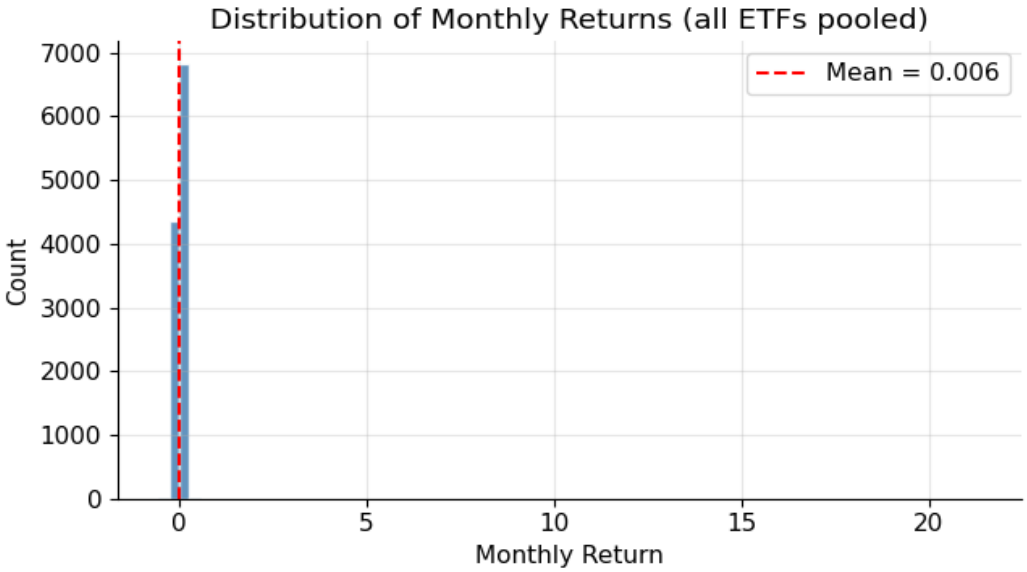
- Assets in uptrends may be genuinely riskier. Momentum profits could be compensation for bearing a hidden systematic risk factor.
- Prediction: controlling for the correct risk factor makes momentum returns disappear. Paper tests this — momentum survives, but debate continues.

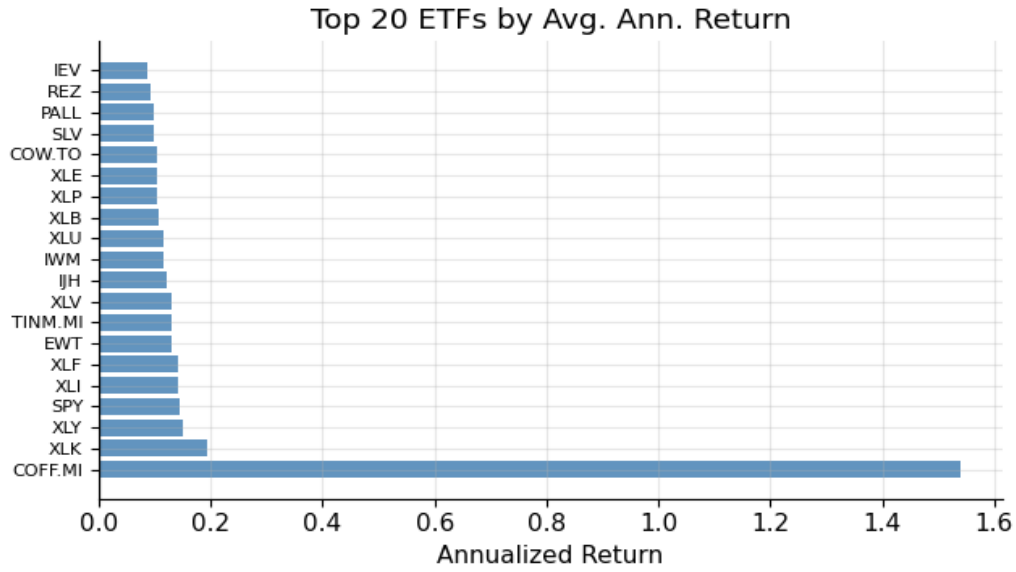
Paper's Universe and Key Numbers

- 58 liquid futures (stocks, bonds, commodities, currencies), Jan 1985–Dec 2009 (25 years). TSMOM (12m lookback) Sharpe ≈ 1.0. Effect holds across all asset classes and all subperiods.

Data: 66 ETFs, February 2012 – March 2026

66	170	7.61%	72.73%
ETFs (4 asset classes)	Monthly observations	Ann. mean return (pooled)	Ann. vol (pooled)





Left: Monthly return distribution — fat tails (COVID, oil crashes). Mean = +0.63%/mo = +7.61% ann. Right: Top-20 ETFs by ann. return — almost all positive (bull market 2012–2026). Key: no sustained bear markets in our sample.

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Building the Trend Signal

EWMA volatility estimation and the 12-month sign rule

Measuring Volatility: EWMA and the 40% Target

Ex-Ante EWMA Volatility Estimate

$\sigma_t^2 = 12 \cdot \Sigma\,(1-\delta)\cdot\delta^{\mathrm{i}} \cdot r^2_{\{m-1-i\}}$ (annualized via ×12)

Monthly approximation: EWMA(r², COM=3 months) ≈ paper's 60 trading-day daily EWMA

vol[t] = vol_estimate.shift(1) → uses only data through t-1. No look-ahead. ✓

Why Volatility-Scale Positions?

- Without scaling: oil (40% vol) overwhelms bonds (15% vol) — all risk from one asset.
- With scaling: each asset contributes EQUAL ex-ante risk. The 40% target means oil gets half the dollar allocation of bonds.
- Bond ETF (15% vol): weight = 40%/15% = 2.67. Oil ETF (40% vol): weight = 1.00. Both contribute ~same risk.

Highest-Vol ETFs in Our Universe

- UNG (natural gas): 39.6% median vol
- URA (uranium): 31.0%
- TUR (Turkey equities): 28.6%
- EWZ (Brazil equities): 28.3%
- USO (crude oil): 26.7%
- SLV (silver): 23.8%

The Trend Signal: 12-Month Sign Rule

TSMOM Signal (Paper main specification)

$s_{\{i,t\}}$ = sign(12-month rolling cumulative return), shifted 1 period — no look-ahead ✓
+1 (went up → BUY / go LONG) or −1 (went down → SELL SHORT)

Design Choices

- Why the SIGN (not the magnitude)? Direction is what matters — a 1% and a 30% gain both give signal +1. Robust to extreme months dominating the signal.
- Why 12 months? Tested many windows; 12 months works well across all asset classes. Matches the well-known equity momentum horizon.
- Fraction long (pooled avg): 62.6% — above 50% due to bull market period.

The Lagging Nature of the Signal

- After a crash, the 12-month window takes months to 'forget' prior gains before turning negative.
- COVID 2020: market fell −34% in 5 weeks, but the 12-month signal stayed LONG for most of 2020 (prior 11 months of gains outweighed one bad month).
- This lag misses sharp reversals — but avoids whipsaw in noisy sideways markets.

3

Statistical Validation

Pooled predictive regressions h = 1–24 months

Does Past Performance Predict Future Returns?

Pooled Predictive Regression (for each horizon h = 1..24)

$$r_{\{i,t\}} \,/\, \sigma_{\{i,t-1\}} \;=\; \alpha \;+\; \beta_h \cdot sign(h - month\ past\ return) \;+\; \varepsilon_{\{i,t\}}$$

`Pool all 66 ETFs × 170 months. HC0 robust SEs. β > 0 = past trend → future outperformer.`

What is a t-statistic?

- t = estimate / standard error. Measures how confident we are the effect is real, not noise.
- |t| > 2.0: ~95% confident the effect is real (5% significance level).
- |t| < 2.0: cannot rule out the effect is random chance.
- |t| = 38 looks impressive BUT with 11,000+ pooled obs, tiny effects produce huge t-stats. Always check β too.

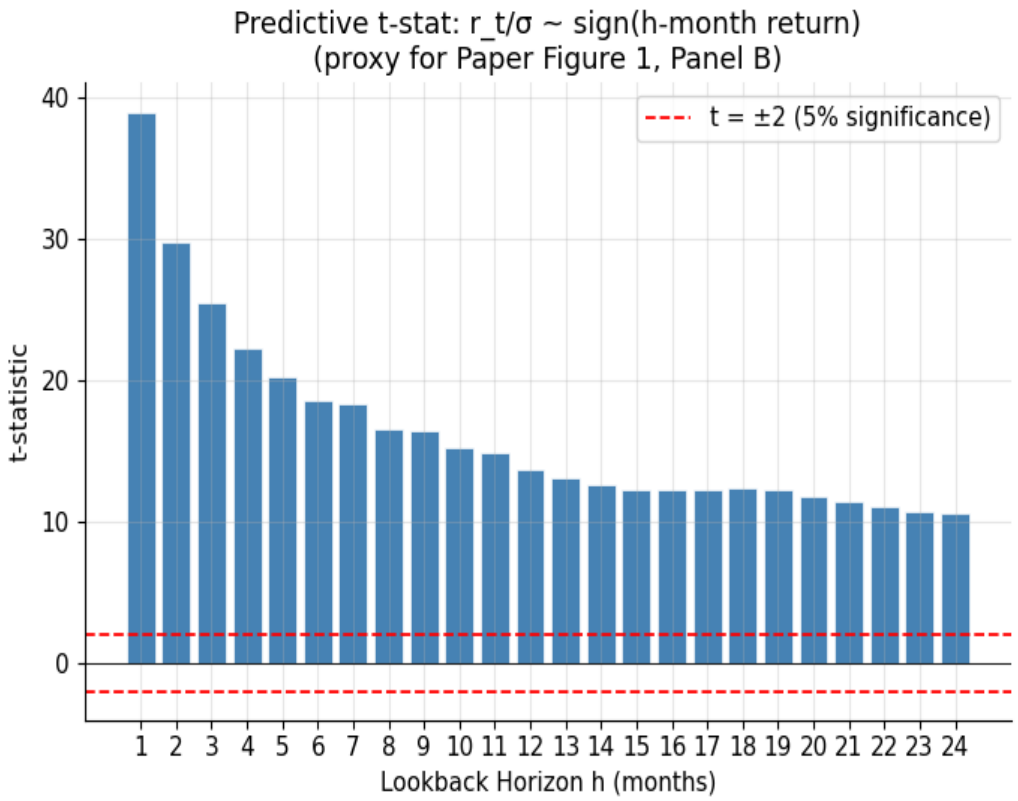
Key Caveats for Our Sample

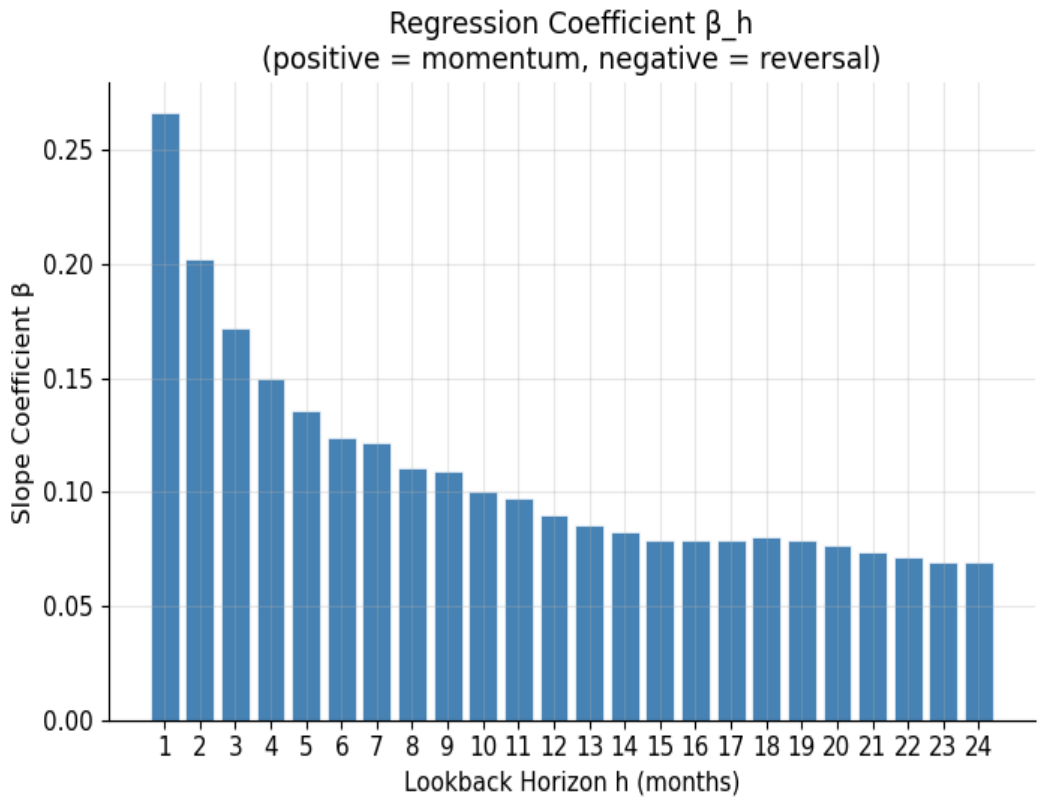
- Pooling 66 correlated ETFs inflates N — they are not 66 independent signals per month.
- Bull market: if everything trends up, past winners and future winners are the same set — not necessarily because of momentum, but general market trend.
- High t-stats confirm statistical detects ability, not large economic effect.

β coefficient is the right metric of effect size

- For h=12: β ≈ 0.09 means the vol-scaled return of a past-winner is 0.09 higher than a past-loser. In practice: moving from the bottom to top decile of past returns → roughly 0.9% higher next-month vol-scaled return. Economically real but modest.

Predictive t-Statistics by Formation Horizon h = 1–24 Months





Left: t-statistics by horizon — ALL 24 bars exceed t=2 (red dashed), with h=1 reaching t≈39. Right: β coefficients — declining from 0.27 (h=1) to 0.07 (h=24). Effect is statistically detectable at all horizons but economically modest. Caution: t-stats inflated by pooled correlated ETFs in sustained bull market.

Selected Results — All 24 Horizons Significant

38.99	13.72	0.090	24 / 24
t-stat at h=1 month	t-stat at h=12 months	β coeff. at h=12	Significant horizons

Horizon h	t-statistic	β coefficient	Interpretation
h=1	38.99	0.2662	Very strong — partly bull market / correlated ETF artifact
h=3	25.46	0.1722	Strong short-term momentum
h=6	18.51	0.1235	Medium-term momentum — consistent
h=12	13.72	0.0900	Paper's flagship — confirmed, modest effect
h=18	12.39	0.0803	Momentum slowly fading at longer horizons
h=24	10.62	0.0690	Still positive — no reversal yet at 2 years

4 Position Sizing

Equal risk contribution and the 40% volatility target

Building the Portfolio: Equal Risk Contribution

Portfolio Position Weight (Paper page 232)

$w_{\{i,t\}} = (40\% \,/\, \sigma_{\{i,t-1\}}) \times s_{\{i,t\}}$ `[direction × vol-scaled size]`

$r_t^{TSMOM} = (1/N_t) \times \Sigma_i \; w_{\{i,t\}} \times r_{\{i,t\}}$ `[averaged across N_t assets with valid signals]`

How the Scaling Works

- Bond ETF (15% vol): weight = 40%/15% = 2.67 — large position
- Equity ETF (25% vol): weight = 40%/25% = 1.60 — medium position
- Oil ETF (40% vol): weight = 40%/40% = 1.00 — small position
- All three contribute roughly the same ex-ante risk to the portfolio.

Portfolio Facts

- Avg long positions per month: 38.4 of 66 ETFs
- Avg short positions per month: 23.0 of 66 ETFs
- Avg |net weight|: 1.70 (slightly net long — bull market bias)
- In bull market: net exposure tilts positive; short leg barely fires.

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Backtesting

Performance summary, cumulative returns, and drawdowns

Backtest Performance Summary (2013–2026)

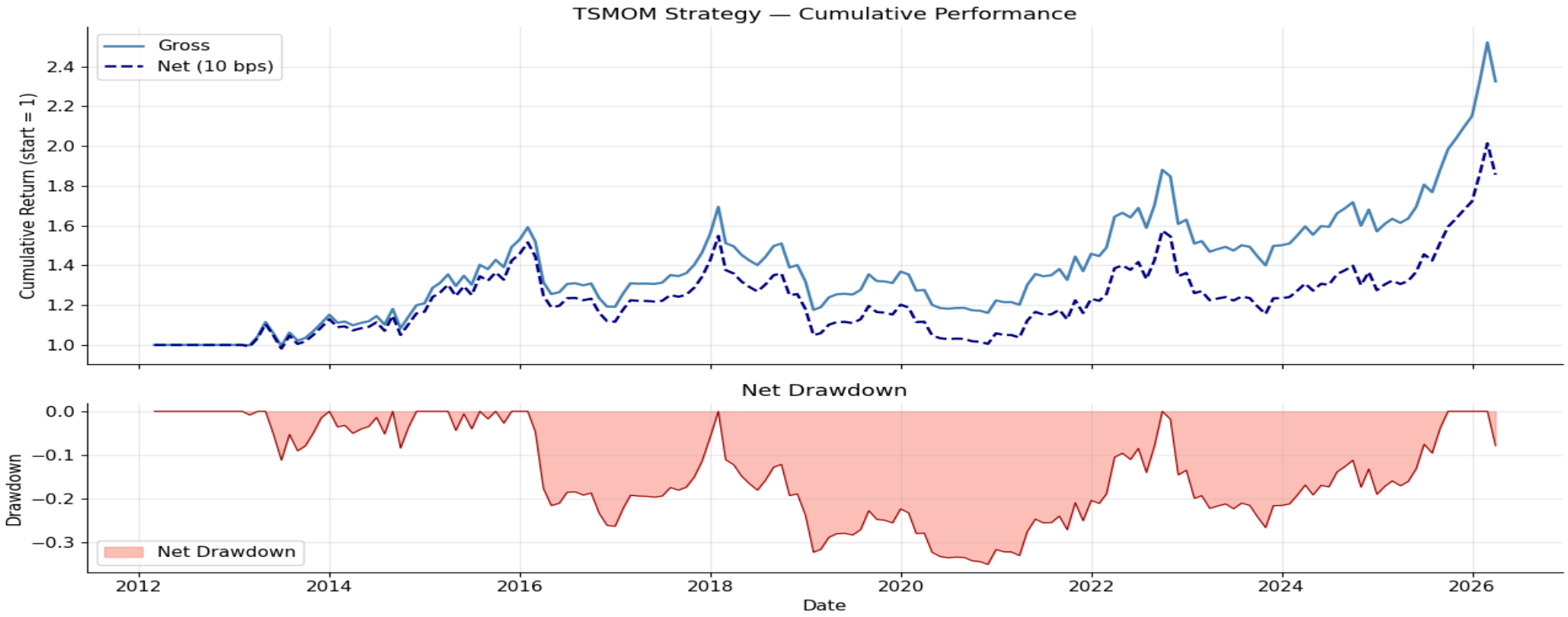
0.48	0.37	−35.0%	7.1%
Gross Sharpe Ratio	Net Sharpe (10bps)	Max Drawdown (net)	Gross Ann. Return

Metric	TSMOM Gross	TSMOM Net (10bps)	Paper Benchmark
Ann. Return	7.1%	5.5%	~12% (futures, excess)
Ann. Volatility	14.9%	14.9%	~12%
Sharpe Ratio	0.48	0.37	~1.0+
Sortino Ratio	0.64	0.49	~1.5+
CAGR	6.1%	4.5%	~11%

Why Is Our Sharpe ~0.37 vs Paper's ~1.0?

- Bull market bias: short positions in rising assets drag returns. Shorter sample (14 vs 25 yrs): wider estimation uncertainty. ETFs more correlated than paper's diverse futures universe.

Cumulative Returns and Drawdown (2013–2026)



Top: Gross vs net cumulative return. Gap = cost drag. 2022 rate-shock creates a strong rally for TSMOM (bonds and equities in clear downtrends). Bottom: Drawdown — worst trough ≈ −35%, recovering slowly. COVID 2020 shows brief spike then loss (quick reversal hurts lagged signal).

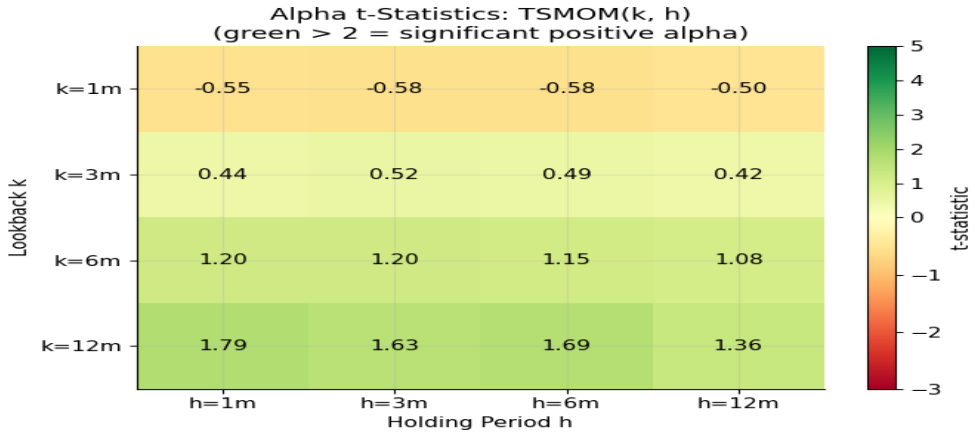
6

Comparing Results to the Paper

Alpha grid, TSMOM vs. passive, and the straddle property

Alpha t-Statistics: (Lookback k, Holding h) Grid

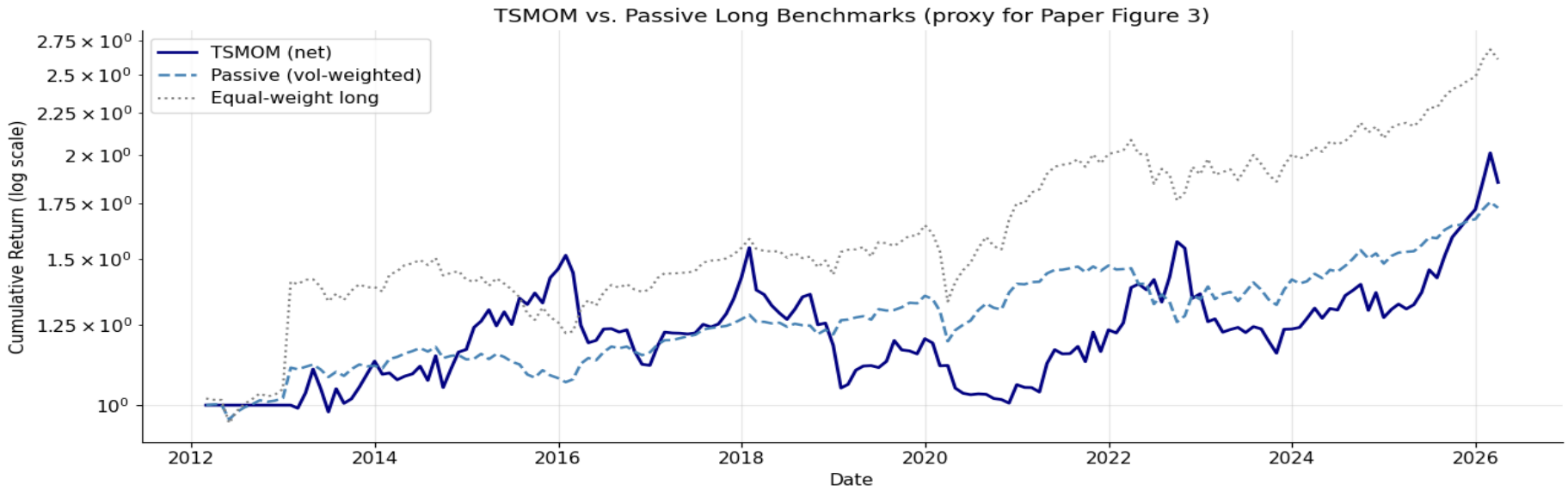
Lookback k	Hold h=1m	Hold h=3m	Hold h=6m	Hold h=12m	Verdict
k=1m	−0.55	−0.58	−0.58	−0.50	Negative — short-term reversal
k=3m	0.44	0.52	0.49	0.42	Positive — not significant
k=6m	1.20	1.20	1.15	1.08	Positive — below t=2 threshold
k=12m	**1.79**	**1.63**	**1.69**	**1.36**	**Best — borderline significant**



Comparison to Paper

- Paper: ALL 16 cells green with t > 3.
- Ours: k=1m row is NEGATIVE. k=12m is best but t < 2.
- Reason: 14-year bull market sample + no factor controls.
- With only 170 months, genuine Sharpe 0.5 → expected t ≈ 1.5–2.0.

TSMOM vs. Passive Long Benchmarks



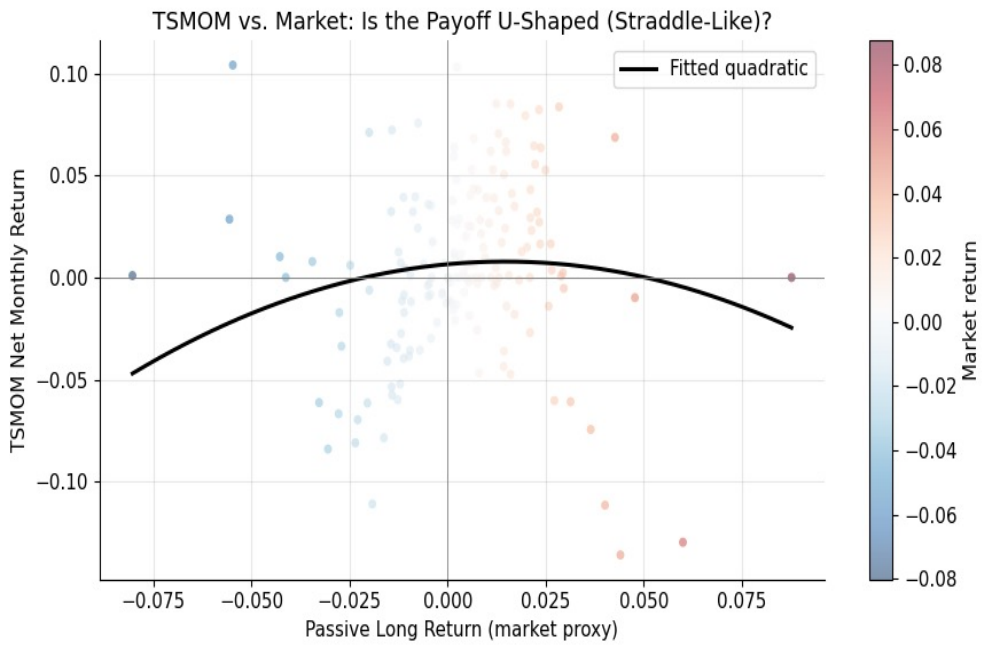
Log-scale: equal-weight long benchmark (gray dotted) outperforms TSMOM (navy) over the full bull market period. Passive vol-weighted: lower absolute return but HIGHER Sharpe (0.59 vs 0.37) due to lower volatility.

Strategy	Ann. Return	Ann. Vol	Sharpe	Max DD
TSMOM (net)	5.5%	14.9%	0.37	−35.0%
Passive vol-weighted	4.1%	7.0%	0.59	−14.5%
Equal-weight long	7.6%	13.4%	0.57	−19.1%

Does TSMOM Profit in Both Crashes AND Rallies?

Factor Regression

- Model: $r^{TSMOM} = \alpha + \beta_1 \cdot MKT + \beta_2 \cdot MKT^2$
- $\beta_2 > 0$: U-shaped (straddle-like)
- $\beta_2 < 0$: hump-shaped (concave)
- Our results:
- $\alpha = 0.0065$ (t=2.05) ✅
- $\beta_1 = 0.178$ (t=0.75)
- $\beta_2 = -6.085$ (t=−1.24) ❌
- $R^2 = 0.023$



Paper vs Ours

- Paper: $\beta_2 > 0$ (straddle). Three major bear markets revealed short-side profits.
- Ours: $\beta_2 < 0$ (concave). Only COVID 2020 — too brief (5 weeks) for sustained short-side gain.
- Alpha: 7.8% ann. significant (t=2.05) — consistent with paper.

Why Our β_2 is Negative

- The straddle shows up during SUSTAINED bear markets (2000–02 dot-com bust, 2008–09 GFC). Without prolonged downtrends, the short side barely contributes. Our sample simply lacks those environments.

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Robustness Checks

Subperiod stability, cost sensitivity, and lookback sensitivity

Subperiod Stability and Cost Sensitivity

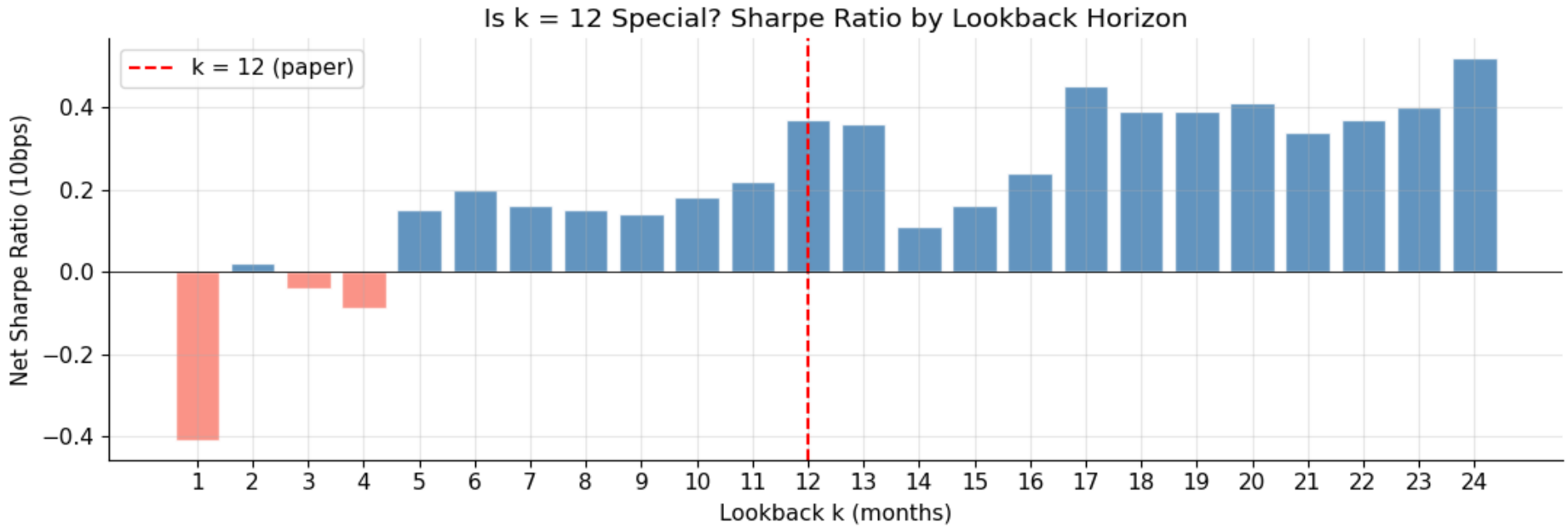
Period	Ann. Return	Sharpe	Max DD
Full (2013–2026)	5.5%	0.37	−35.0%
First half (2012–2019)	1.9%	0.13	−32.2%
Second half (2019–2026)	9.0%	0.61	−26.6%

First Half vs Second Half

- First half (2012–2019): Sharpe only 0.13 — quiet bull market, short side consistently loses. Second half (2019–2026): Sharpe 0.61 — COVID, 2022 rate-shock, AI rally give more regime diversity.

Cost (bps)	Sharpe	CAGR	Net Ann. Return
0	0.480	6.1%	7.1%
5	0.420	5.3%	6.3%
10	0.370	4.5%	5.5%
25	0.210	2.0%	3.1%

Is k = 12 Months Truly Special?



Net Sharpe by lookback k (1–24 months). Paper's k=12 (red dashed). Short lookbacks k=1–4 are negative (reversal). Longer lookbacks k=17–24 look better in this specific sample — but that's in-sample selection. Paper chose k=12 based on economic reasoning, not by cherry-picking from the data.

Why k=12 Is Justified Despite Noise

- With 24 tests, 1–2 look good by chance alone (multiple comparison problem). The paper chose k=12 based on ECONOMIC reasoning (≈1 year for trends to establish). Cross-asset robustness is what gives the choice credibility — not data-mining.

8

Honest Assessment

What replicates, what doesn't, and why that is itself informative

Honest Assessment: Paper vs. Our Replication

Result	Paper	Our Notebook	Main Reason
Predictive t-stats	t = 3–5 for h=1..12m	t > 10 for ALL horizons	Inflated by pooling correlated ETFs in bull market
Alpha grid	Mostly t > 3	Mostly t < 2	Short sample; no factor controls; bull market
TSMOM vs passive	TSMOM clearly wins	Mixed — passive often wins	Bull market favors long-only; short side hurts
Straddle (β_2)	Positive, significant	$\beta_2 < 0$ (likely negative)	No sustained bear market in our 14-year sample
Subperiod stability	Robust across cycles	First half weak, second half ok	Regime-dependent; 14 years too short
Gross Sharpe	~1.0+	~0.48	All of the above combined
Net Sharpe (10bps)	~0.9+	~0.37	Cost drag adds to the gap

Why Our Results Differ: Six Structural Reasons

Factor	Paper	Our Notebook	Impact
Sample period	Jan 1985–Dec 2009 (25 yrs)	Feb 2012–Mar 2026 (~14 yrs)	Less data; bull-market biases results
Universe	58 liquid futures (diverse)	66 ETFs (more correlated)	Less diversification; weaker signal
Return definition	Excess over T-bill	Total returns (no T-bill)	Slightly inflated raw return levels
Volatility model	Daily EWMA, 60-day	Monthly EWMA, 3-month	Less precise estimates
Market regime	Three major bear markets	One crash (COVID 2020)	Short side barely contributes

The Core Message

- The most important difference: the TIME PERIOD. The paper included three major multi-year bear markets where the short side earned its keep. Our 14-year sample has only one brief crash. This does NOT invalidate the academic finding — it shows how period-specific samples produce different results.

Key Takeaways

1. Time series momentum buys assets with positive 12-month past returns and shorts those with negative — using their OWN history. Cross-sectional momentum ranks assets against each other. The two can give OPPOSITE signals.

2. The 12-month sign rule is the signal. EWMA volatility (COM=3 months) scales positions so every asset contributes equal ex-ante risk. Both are shifted by 1 period — no look-ahead bias.

3. Predictive regressions confirm positive momentum at all 24 tested horizons. But high t-stats (t=38 at h=1) partly reflect 11,000+ pooled correlated ETFs in a bull market. Always examine β, not just t.

4. Our gross Sharpe of 0.48 and net Sharpe of 0.37 are well below the paper's ~1.0 — primarily because our 2012–2026 sample is a bull market with only one brief crash (COVID 2020).

5. The straddle property (profits in big ups AND big downs) is the paper's most interesting finding — but requires sustained bear markets to show up. COVID lasted 5 weeks before full recovery. We could not replicate it.

6. Statistical power: 170 monthly observations gives a 95% CI of roughly ±0.15 around any Sharpe estimate. Do not over-interpret point estimates from short samples.

7. Context is everything: the same strategy looks completely different across regimes. Always ask: 'Does my sample include the environments where this strategy earns its return?'

Discussion Questions

1. The strategy uses only the SIGN of past returns, not the magnitude. Would a stronger signal come from using the actual return size? What could go wrong — and what does this tell us about the role of robustness in factor design?

2. Our first half (2012–2019) had a Sharpe of only 0.13. Can you design a modification that would help the strategy in bull markets without sacrificing its advantage in crashes? What tradeoffs does that modification create?

3. We tested 24 lookback windows and then commented on which performed best. Is that a valid way to choose k? How would you design a proper out-of-sample test to avoid data-snooping bias?

4. If you held TSMOM alongside a standard 60/40 equity-bond portfolio, would you expect the combined portfolio to have higher or lower risk-adjusted returns? What does your answer depend on?

5. The straddle property ($\beta_2 > 0$) is the paper's most interesting prediction, but we couldn't replicate it. What specific events from 2012–2026 could have given us a better chance of seeing it? Does its absence mean it doesn't exist?

The Technical Recipe at a Glance

Component	Formula / Implementation
Monthly return	$r_t = (P_t - P_{\{t-1\}}) / P_{\{t-1\}}$ `(simple return, not excess)`
EWMA volatility	$\sigma_t^2 = 12 \cdot$ `EWMA(r², COM=3 months), then .shift(1) — no look-ahead`
Trend signal	$s_{\{i,t\}} =$ `sign(12-month rolling sum of r), shifted 1 period`
Position weight	$w_{\{i,t\}} = (40\% / \sigma_{\{i,t-1\}}) \times s_{\{i,t\}}$ *then average over* N_t *assets*
Gross return	$r_t^{TSMOM} = \Sigma_i\, w_{\{i,t\}} \times r_{\{i,t\}} / N_t$
Transaction cost	$TC_t = 0.0010 \times \Sigma_i\, \lvert w_{\{i,t\}} - w_{\{i,t-1\}} \rvert$ `(10 bps per unit turnover)`
Net return	$r_t^{net} = r_t^{gross} - TC_t$

Results at a Glance

- Gross Sharpe 0.48 | Net Sharpe 0.37 | CAGR 4.5% net | Max DD −35.0% | First half Sharpe 0.13 | Second half Sharpe 0.61 | Paper target: Sharpe ~1.0

References & Further Reading

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